

Fingerprint Enhancement - Batch Experiment Summary

Method	MSE_mean	PSNR_mean	SSIM_mean	Coherence_mean	Match_mean	Runtime_mean
Global HE	0.0268	16.0136	0.6075	0.675	15.0694	0.0061
Hybrid	0.0447	13.5796	0.5888	0.6938	4.4583	0.4778
Wiener + CLAHE + Unsharp	0.0467	13.408	0.602	0.6957	4.25	0.0098
M1 Wiener + CLAHE	0.0489	13.2037	0.5975	0.708	4.2083	0.0089
M1 Median + CLAHE	0.067	11.8726	0.5567	0.6537	4.1667	0.0088
M1 CLAHE	0.0871	10.7384	0.5028	0.5845	3.5972	0.0059
Gabor	0.0747	11.3246	0.4334	0.5984	2.5	0.4819
M1 CLAHE + Unsharp	0.0933	10.4387	0.4634	0.5301	2.5972	0.007
Degraded input	0.0312	15.5893	0.5196	0.5953	16.8889	0.0

Dataset mode: SOCOFing Real images with controlled degradation

Example processing pipeline

Clean reference



Low-quality input



Member 1: restoration + CLAHE



Member 2: Gabor



Member 3: Restored ridges

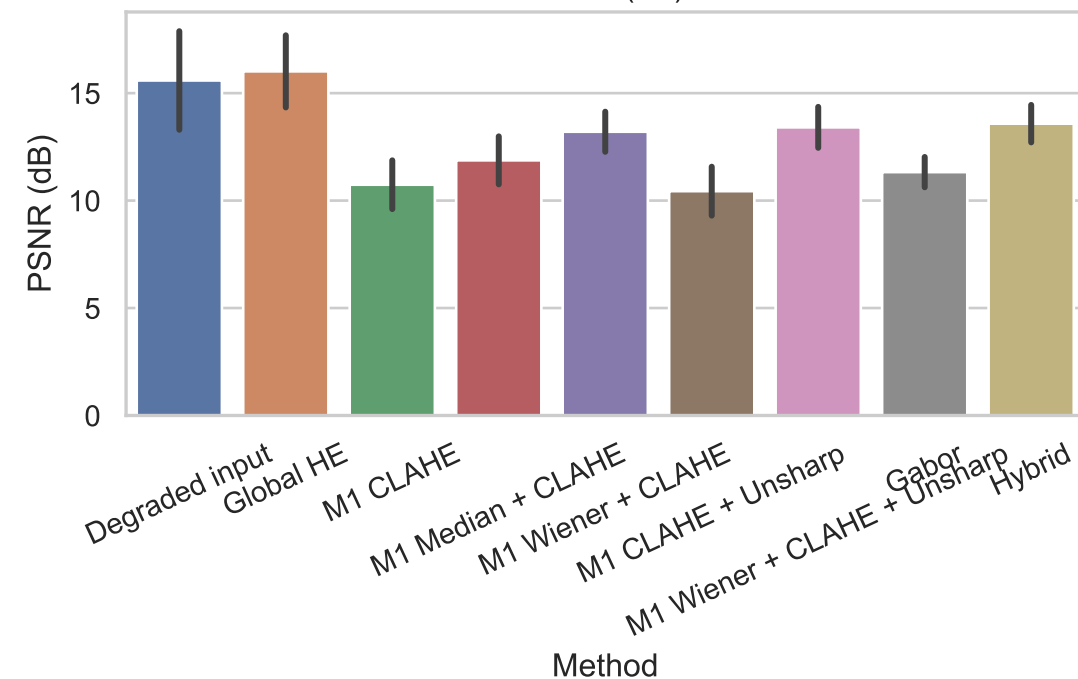


Member 4: Skeleton

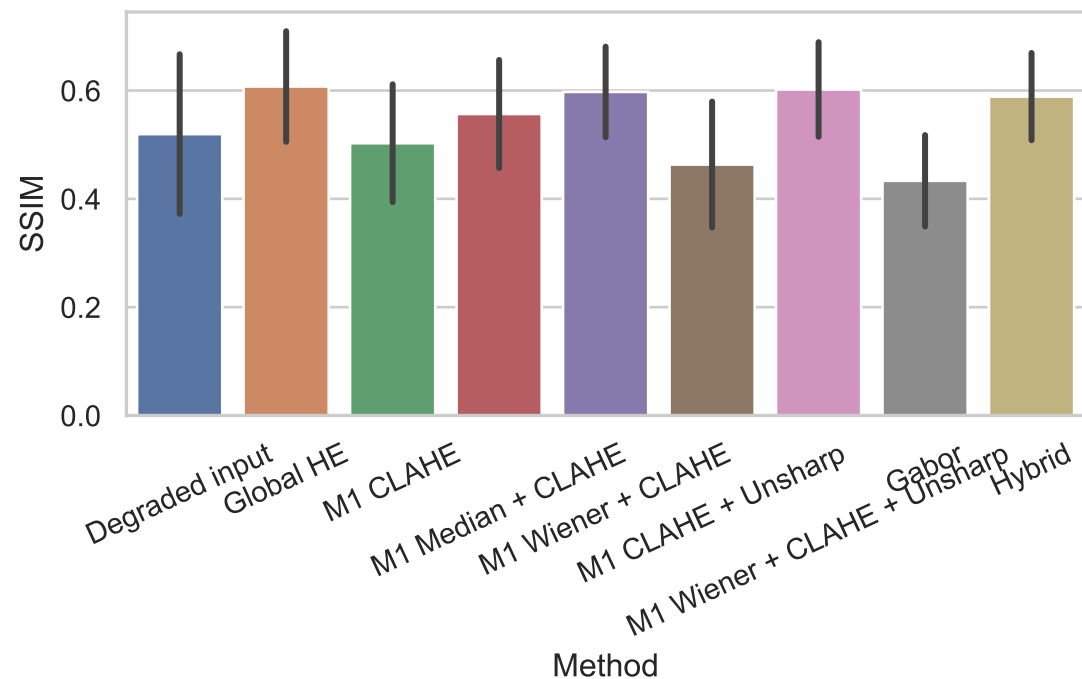


Mean performance with standard deviation

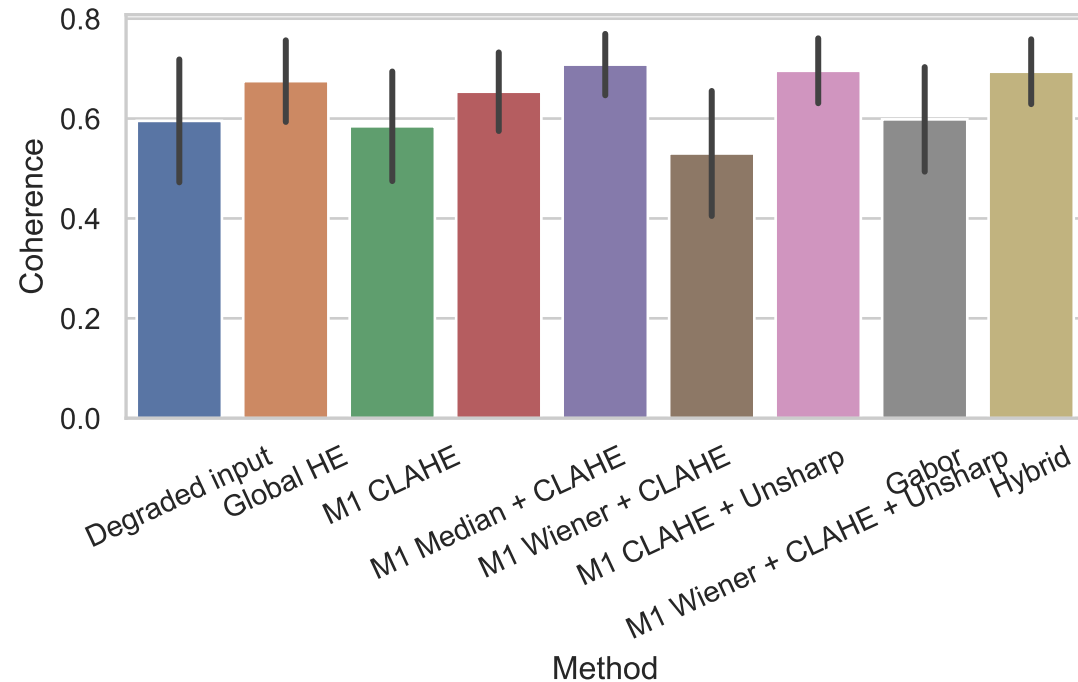
PSNR (dB)



SSIM



Coherence



Match score (%)

